

Nima Jafari

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Summary: Full-Stack Software Engineer with AI/ML, LLM, RAG, and agentic workflow experience across web, mobile, backend, and database-backed applications. Experienced with AI-assisted development tools, reviewing AI-generated code, and building systems where deterministic backend logic, user-scoped data, and auditability constrain LLM behavior. Awarded best paper in computational sciences and mathematics at Stanford University SRC Conference 2026.

Skills

Programming: Python, Java, C++, Swift, SwiftUI, JavaScript/TypeScript, HTML/CSS, SQL, React, Solid.js, React Native

AI/ML & LLM Systems: Reinforcement learning (PPO, Stable-Baselines3), deep learning (PyTorch, TensorFlow/Keras), supervised learning, clustering, time-series forecasting, RAG, graph-grounded retrieval, agentic workflow orchestration, LLM application pipelines, prompt engineering, LLM output validation, AI-generated code review/evaluation, LangChain, Ollama, Jupyter

AI Coding & Developer Tools: Cursor, Claude Code, OpenAI Codex, assisted prototyping, code review

Frameworks & Tools: FastAPI, Node.js/Express, Next.js, React, Tailwind CSS, Electron, Playwright, scikit-learn, NumPy, Pandas, Docker, Docker Compose, Git, CI/CD, Alembic, pytest, Kubernetes

Cloud, Databases & Infrastructure: AWS (RDS, EC2, S3), Azure, PostgreSQL, MySQL, SQLite, Neo4j, Redis, MinIO, Qdrant, REST APIs, data pipelines, microservices, system design, cloud deployment, Git/GitHub

Experience

Full-Stack Software Engineer | Stealth

Aug 2025 – June 2026

- Engineered backend systems and REST APIs for AI-driven applications, enabling reliable communication between frontend clients, business logic, and data services.
- Used AI coding tools including GitHub Copilot, Cursor, Claude Code, and OpenAI Codex for prototyping, debugging, and refactoring while reviewing AI-generated code for correctness, maintainability, security, and integration with existing application logic.
- Built end-to-end full-stack workflows connecting frontend interfaces to backend services, including API integration, request handling, and database-driven functionality.
- Designed and integrated hybrid RAG and LLM-powered application pipelines with retrieval, ranking, and backend orchestration to improve answer quality, domain relevance, and context-aware outputs.
- Developed finance-oriented AI features for analysis, recommendation, and workflow automation, transforming structured and unstructured data into user-facing insights.
- Owned full-stack implementation across backend architecture, API design, and frontend integration, debugging cross-layer issues and improving system reliability.

Full-Stack iOS Engineer | SideShift NYC

May 2026 – May 2026

- Contributed across SwiftUI mobile development, backend engineering, web application features, and subscription growth initiatives.
- Improved mobile UI/UX quality and resolved interface issues in SwiftUI.
- Reduced unnecessary cloud database/API usage by optimizing backend data access and mobile caching behavior.
- Built and refined backend-supported web application features for platform workflows.
- Supported product discussions related to subscription conversion and premium feature adoption.
- Collaborated through feedback cycles to deliver production-oriented fixes, feature improvements, and implementation notes.

Machine Learning Engineer | CSUN

Jan 2023 – July 2026

- Conducted machine learning research across NSF-, AFOSR-, and DARPA-supported projects, contributing to experiments, proposal development, and technical writing for funded research efforts.
- Built custom robotics simulation environments and reproducible logging/benchmarking infrastructure that supported systematic reinforcement learning experimentation and analysis.
- Developed an RL-to-symbolic analysis pipeline with a custom Hasse-diagram clustering method that recovered 2 winning strategies covering 100% of 125 successful episodes.

- Evaluated robustness under 10% corruption, showing that the proposed Hasse-clustering approach preserved full coverage while DBSCAN and hierarchical clustering underperformed or required tuning to approach similar consensus.
- Developed computer vision pipelines for sports-motion analysis using YOLO pose and other pose-estimation encoders, and trained CNNs on sports video data for successful figure skating jump-type detection.
- Co-authored 1 arXiv preprint and contributed to 2 ongoing research projects.
- Presented findings at CSUN research events, the AFOSR 2025 Computational Cognition & Machine Intelligence Program Review in Washington, D.C., Stanford SRC 2026, and invited academic talks in the U.S. and abroad.

Python Software Developer | MISAN Robotic Foundation

Jun 2020 – Jun 2021

- Supported robotics projects for firefighting and emergency-response applications.
- Contributed to programming, hardware assembly, and mechatronic integration tasks in a hands-on development environment.
- Assisted with prototype development and testing for robotics systems in safety-critical use cases.

Selected Projects

AgenticAI (AI-Assisted Refund Support Workflow)

- Built an agentic, backend-orchestrated refund support system where a Streamlit chat frontend connects to a FastAPI backend that extracts refund details, validates customer/order ownership, evaluates policy, and returns customer-facing responses. (Python, FastAPI, Streamlit, Pydantic, SQLite)
- Designed a deterministic PolicyEngine as the sole refund-decision authority, keeping the LLM limited to missing-information prompts, blocked-state responses, and final customer-facing explanations rather than financial decision-making. (Ollama, OpenAI SDK, Anthropic SDK, prompt engineering)
- Implemented auditable safety controls including session-scoped, single-use, type-checked decision tokens; SessionGuard terminal-state locking; trace/tool-call persistence; provider fallback logging; and tests for safety boundaries and LLM fallback behavior. (pytest, Ruff, SQLite, YAML/Markdown policy-as-code)

PocketPilot (Personal Finance & AI Insights Platform)

- Built a full-stack personal finance application that lets users import transactions, automatically categorize spending, create budgets and savings goals, and analyze financial behavior through interactive dashboards. (React Native, TypeScript, Vite, Tailwind CSS, Firebase, iOS development, Rust, Solid.js)
- Designed a monorepo architecture with shared business-logic and service layers, secure user-scoped data access, and backend workflows for transaction import, budget tracking, and goal-progress visualization. (Firestore, Firebase Auth, Cloud Functions, Node.js)
- Integrated a local retrieval-augmented AI pipeline using Qdrant and Ollama to support user-specific financial Q&A and explainable spending insights. (Qdrant, Ollama, Docker)

Education

California State University, Northridge (CSUN)

Fall 2022 – Spring 2026

B.S. in Computer Science | Cumulative GPA: 3.98/4.00

Honors & Awards

- **CSUN CSE Department Scholarships:** Guerrera Endowed Scholarship (2023), Engineering Merit Scholarship (2024), Trustee Steven G. Stepanek Endowed Scholarship (2025).
- **Iranian American Women's Foundation (IAWF):** Scholarship Awardee (2024, 2025); active mentee and volunteer.
- **Dean's List:** Fall 2022; Spring 2023; Fall 2023; Spring 2024; Fall 2024; Spring 2025; Fall 2025 (ongoing).

Leadership & Service

Alpha Lambda Delta Honor Society

2023 – Present

- Co-founded the chapter; served as Treasurer for the 2024 academic year.

Community Engagement

Ongoing

- Volunteer: American Red Cross (LA fire), United Methodist Church of Northridge (Wednesday labor crew), The Church at Rocky Peak (Café; Young Adults worship band).
- Active member of the USTA and City of Sylmar tennis team.